SAECOIN

APPLICATION FLOW SPECIFICATION

Project: SAECOIN
Ticker: SAC (Provisional)
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1. DOCUMENT PURPOSE

This document defines the complete user journeys, application navigation, screen transitions, and functional workflows of the SAECOIN software ecosystem.

All applications must follow the same blockchain protocol and user experience principles.

The applications must communicate with SAECOIN nodes to retrieve blockchain information and broadcast transactions.

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2. SOFTWARE COMPONENTS

SAECOIN will consist of five major software components.

Component| Platform| Purpose
SAECOIN Wallet| Android| Mobile cryptocurrency wallet
SAECOIN Wallet| Windows| Desktop cryptocurrency wallet
SAECOIN Node| Windows/Linux| Blockchain node operation
SAECOIN Miner| Windows/Linux| CPU and GPU mining
SAECOIN Explorer| Web| Public blockchain exploration

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3. ANDROID WALLET APPLICATION FLOW

3.1 Application Launch

When the user opens SAECOIN Wallet:

Splash Screen
↓
Application Initialization
↓

Check Wallet Configuration
↓
Is Wallet Already Created?

YES → Wallet Unlock Screen

NO → Welcome Screen

3.2 Welcome Screen

The Welcome Screen provides two primary options.

1. Create New Wallet
2. Import Existing Wallet

Additional options:

- About SAECOIN
- Security Information
- Terms of Use

Navigation:

Welcome Screen
├── Create Wallet
└── Import Wallet

3.3 Create New Wallet

User Flow:

Welcome Screen
↓
Create Wallet
↓
Security Warning
↓
Generate Wallet
↓
Generate Recovery Phrase
↓
Display Recovery Phrase
↓
User Confirms Recovery Phrase

↓
Set Wallet Password
↓
Wallet Successfully Created
↓
Dashboard

Requirements:

- Generate cryptographically secure wallet keys.
- Generate a recovery phrase using an established wallet standard.
- Display recovery phrase securely.
- Require recovery phrase confirmation.
- Encrypt locally stored private keys.
- Never transmit recovery phrases to external servers.

The recovery phrase must be shown only during the secure wallet creation or recovery process.

3.4 Import Existing Wallet

User Flow:

Welcome Screen
↓
Import Wallet
↓
Select Import Method
↓
Enter Recovery Phrase
↓
Validate Recovery Phrase
↓
Restore Wallet
↓
Set New Wallet Password
↓
Wallet Imported Successfully
↓
Dashboard

Import methods may include:

- Recovery Phrase.
- Supported Wallet Backup File.

Unsupported wallet formats must be rejected with a clear explanation.

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4. ANDROID WALLET DASHBOARD

The Dashboard is the primary screen.

It must display:

- Total SAC Balance.
- Available Balance.
- Pending Balance.
- Wallet Address.
- Latest Transactions.
- Network Status.
- Current Block Height.
- Synchronization Status.

Primary navigation:

Dashboard
├── Send
├── Receive
├── Transactions
├── Wallet Management
├── Network
└── Settings

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5. SEND SAC FLOW

User Flow:

Dashboard
↓
Tap Send
↓
Enter Recipient Address
↓
Enter SAC Amount
↓
Display Estimated Transaction Fee
↓
Validate Recipient Address
↓
Validate Available Balance

↓
Review Transaction
↓
Confirm Transaction
↓
Authenticate Wallet
↓
Sign Transaction Locally
↓
Broadcast Transaction
↓
Transaction Submitted
↓
Transaction Details

Requirements:

- Recipient address must be validated.
- Amount must be greater than zero.
- Available balance must cover amount and fee.
- Transaction fee must be displayed before confirmation.
- Transaction must be signed locally.
- Private keys must never leave the wallet.
- Transaction ID must be displayed after successful submission.

Error conditions:

- Invalid Address.
- Insufficient Balance.
- Insufficient Fee.
- Network Unavailable.
- Transaction Rejected.
- Wallet Locked.

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6. RECEIVE SAC FLOW

User Flow:

Dashboard
↓
Tap Receive
↓
Display Wallet Address
↓
Generate QR Code
↓

Copy Address
↓
Share Address
↓
Wait for Incoming Transaction
↓
Detect Transaction
↓
Display Pending Transaction
↓
Wait for Confirmations
↓
Update Balance

The application must display confirmation progress.

Example:

Transaction Received

0 Confirmations → Pending

1 Confirmation → Confirmed (1/6)

2 Confirmations → Confirmed (2/6)

6 Confirmations → Recommended Confirmation Target Reached

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7. TRANSACTION HISTORY FLOW

Dashboard
↓
Transactions
↓
Transaction List
↓
Select Transaction
↓
Transaction Details

Transaction Details must display:

- Transaction ID.
- Sender/Inputs.
- Recipient/Outputs.
- Amount.

- Transaction Fee.
- Block Height.
- Timestamp.
- Confirmation Count.
- Transaction Status.

The application must allow users to copy the Transaction ID.

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8. WALLET MANAGEMENT FLOW

Dashboard
↓
Wallet Management

Available options:

1. Create Additional Wallet.
2. Import Wallet.
3. Switch Wallet.
4. Backup Wallet.
5. Restore Wallet.
6. Export Supported Backup.
7. View Public Addresses.

Private key export must require explicit authentication and a security warning.

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9. WALLET SETTINGS

Settings Screen:

- Security Settings.
- Change Wallet Password.
- Biometric Authentication.
- Network Settings.
- Currency Display.
- Notification Settings.
- Backup Information.
- Application Information.
- About SAECOIN.

Users must be able to verify the network they are connected to.

The application must clearly distinguish Mainnet and Testnet.

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10. WINDOWS WALLET APPLICATION FLOW

Windows Wallet will provide the same essential functionality as Android Wallet.

Application Flow:

Launch Application
↓
Initialize Wallet
↓
Existing Wallet?

YES → Unlock Wallet

NO → Welcome Screen

Welcome Screen:

- Create Wallet.
- Import Wallet.
- Restore Backup.

After authentication:

Dashboard
├── Overview
├── Send SAC
├── Receive SAC
├── Transactions
├── Addresses
├── Network Status
└── Settings

Windows Wallet must support secure local key storage.

The wallet must communicate with a verified SAECOIN node.

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11. DESKTOP NODE APPLICATION FLOW

The Desktop Node Application allows users to participate in the SAECOIN network.

11.1 First Launch

Launch Node Application

↓
Initialization
↓
Select Network
↓
Mainnet / Testnet
↓
Configure Data Directory
↓
Initialize Node
↓
Discover Peers
↓
Connect to Network
↓
Synchronize Blockchain
↓
Node Dashboard

11.2 Node Dashboard

The dashboard must display:

- Node Status.
- Blockchain Height.
- Local Block Height.
- Synchronization Progress.
- Connected Peers.
- Network Difficulty.
- Network Hashrate.
- Latest Block.
- Data Storage Usage.

Primary Controls:

- Start Node.
- Stop Node.
- Restart Node.
- View Logs.
- Open Data Directory.
- Configure Network.

11.3 Node Synchronization

Node Flow:

Start Node
↓
Load Local Blockchain
↓
Discover Peers
↓
Request Blockchain Headers
↓
Validate Headers
↓
Download Required Blocks
↓
Validate Blocks
↓
Store Valid Blocks
↓
Continue Synchronization
↓
Fully Synchronized

Invalid blocks must be rejected.

A node must not trust block information merely because it was received from another peer.

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12. COMMAND LINE INTERFACE FLOW

SAECOIN CLI will provide command-line access to node and wallet functionality.

Planned command structure:

Node Commands

- sae-node start
- sae-node stop
- sae-node status
- sae-node peers
- sae-node sync
- sae-node logs

Blockchain Commands

- sae-cli blockchain info
- sae-cli blockchain height
- sae-cli blockchain block

- sae-cli blockchain difficulty

Wallet Commands

- sae-cli wallet create
- sae-cli wallet import
- sae-cli wallet balance
- sae-cli wallet addresses
- sae-cli wallet transactions

Transaction Commands

- sae-cli transaction create
- sae-cli transaction send
- sae-cli transaction status

Mining Commands

- sae-miner start
- sae-miner stop
- sae-miner status
- sae-miner benchmark

Exact syntax and parameter validation will be defined in the CLI implementation.

Sensitive recovery phrases and private keys must not be exposed through ordinary command output or shell history.

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13. MINING SOFTWARE FLOW

SAECOIN Miner will support CPU and compatible GPU devices.

13.1 Miner Launch

Launch Miner
↓
Initialize Mining Software
↓
Detect Hardware
↓
Display Available Devices
↓
Select Mining Mode
↓
Solo Mining / Pool Mining
↓

Configure Wallet Address
↓
Configure Mining Parameters
↓
Start Mining

13.2 CPU Mining Flow

Miner Dashboard
↓
Select CPU
↓
Configure Thread Count
↓
Run Benchmark
↓
Start CPU Mining
↓
Display Hashrate
↓
Monitor Mining Performance

13.3 GPU Mining Flow

Miner Dashboard
↓
Detect Compatible GPU
↓
Select GPU Device
↓
Run GPU Benchmark
↓
Configure Mining Parameters
↓
Start GPU Mining
↓
Display GPU Hashrate
↓
Monitor Temperature and Performance

Hardware compatibility depends on the final mining algorithm and supported drivers.

13.4 Mining Dashboard

The Mining Dashboard must display:

- Mining Status.
- Selected Mining Algorithm.
- CPU Hashrate.
- GPU Hashrate.
- Total Hashrate.
- Mining Mode.
- Current Block Height.
- Network Difficulty.
- Mining Uptime.
- Accepted Shares (Pool Mode).
- Rejected Shares (Pool Mode).
- Estimated Rewards.

Estimated rewards must never be presented as guaranteed earnings.

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14. SOLO MINING FLOW

User Flow:

Open Miner
↓
Select Solo Mining
↓
Connect to Local or Remote Full Node
↓
Enter Mining Wallet Address
↓
Validate Wallet Address
↓
Select Hardware
↓
Start Mining
↓
Search for Valid Block
↓
Find Valid Proof of Work
↓
Submit Candidate Block to Node
↓
Node Validates Block
↓
Block Accepted or Rejected

↓
Display Mining Result

A block reward must be credited only after a valid block is accepted by the network.

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15. POOL MINING FLOW

User Flow:

Open Miner
↓
Select Pool Mining
↓
Enter Mining Pool Address
↓
Enter Worker Details
↓
Enter Payout Address
↓
Connect to Pool
↓
Receive Mining Work
↓
Perform Mining
↓
Submit Shares
↓
Pool Verifies Shares
↓
Calculate Eligible Rewards
↓
Process Payout According to Pool Rules

Pool accounting and payout rules must be transparent.

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16. BLOCK EXPLORER FLOW

The Block Explorer will be accessible through a public website.

Home Page
↓
Search Bar
↓
Select Search Type

Available search types:

- Block Height.
- Block Hash.
- Transaction ID.
- Wallet Address.

16.1 Explorer Homepage

Display:

- Current Block Height.
- Latest Blocks.
- Latest Transactions.
- Network Hashrate.
- Mining Difficulty.
- Circulating Supply.
- Transaction Statistics.
- Network Status.

16.2 Block Details

User Flow:

Homepage
↓
Select Block
↓
Block Details

Display:

- Block Height.
- Block Hash.
- Previous Block Hash.
- Merkle Root.
- Timestamp.
- Miner Information (if available).
- Difficulty.
- Nonce.
- Block Reward.
- Transaction Count.
- Transaction List.

16.3 Transaction Details

User Flow:

Homepage
↓
Search Transaction ID
↓
Transaction Details

Display:

- Transaction ID.
- Inputs.
- Outputs.
- Amounts.
- Fees.
- Block Height.
- Timestamp.
- Confirmation Count.
- Transaction Status.

16.4 Address Details

User Flow:

Homepage
↓
Search Wallet Address
↓
Address Details

Display:

- Public Address.
- Confirmed Balance.
- Pending Balance.
- Transaction History.
- Received Transactions.
- Sent Transactions.

The Explorer must never display private keys or recovery phrases.

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17. NETWORK STATUS FLOW

All applications must provide network information.

Possible states:

- Connecting.
- Connected.
- Synchronizing.
- Synchronized.
- Disconnected.
- Network Error.
- Outdated Protocol Version.

The application must clearly communicate connection problems.

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18. ERROR HANDLING

All applications must provide understandable error messages.

Examples:

Error| User Message
Invalid Address| Please enter a valid SAC address.
Insufficient Balance| Your wallet does not have enough available balance.
Network Error| Unable to connect to the SAECOIN network.
Invalid Transaction| Transaction validation failed.
Wallet Locked| Unlock your wallet to continue.
Sync Error| Blockchain synchronization failed.
Unsupported Network| This application version does not support the selected network.

Sensitive technical information must not be exposed in ordinary user-facing error messages.

Detailed diagnostics may be written to local logs.

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19. APPLICATION SECURITY FLOW

Security must be integrated into every application.

Requirements:

1. Wallet authentication before sensitive operations.

2. Local transaction signing.
3. Encrypted private key storage.
4. Recovery phrase protection.
5. Network identity verification.
6. Secure application updates.
7. Input validation.
8. Safe error handling.
9. No unauthorized remote key access.
10. No hidden transfer or modification of user funds.

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20. NETWORK ENVIRONMENT SEPARATION

SAECOIN must support separate network environments.

Development Network

Used for local development.

Testnet

Used for public testing.

Mainnet

Used for real SAC transactions.

Each network must have:

- Unique Network ID.
- Unique Genesis Block.
- Unique Address Prefix or Network Identification.
- Separate Data Directory.
- Separate Network Configuration.

Wallets must warn users before broadcasting transactions on a different network.

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21. FUTURE ECOSYSTEM FLOW

Future features may include:

- Third-party Token Creation.
- Developer Tools.
- Token Explorer.
- Advanced Mining Analytics.

- Additional Wallet Features.

These features must not delay the initial independent blockchain implementation.

Token creation must be implemented through a separately specified protocol.

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22. APPLICATION FLOW ACCEPTANCE CRITERIA

The application ecosystem will be considered functionally complete when:

1. Users can create wallets.
2. Users can import wallets.
3. Users can receive SAC.
4. Users can send SAC.
5. Users can inspect transactions.
6. Users can run full nodes.
7. Nodes can synchronize.
8. Miners can perform valid Proof of Work.
9. CPU mining works on supported hardware.
10. GPU mining works on supported hardware.
11. Users can inspect public blockchain data.
12. Network environments are properly separated.
13. Private keys remain under user control.
14. Applications reject invalid operations.
15. Applications display accurate network status.

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23. DEVELOPMENT RULE

All applications must follow the official SAECOIN Blockchain Protocol.

Application interfaces must not independently determine:

- Block validity.
- Transaction validity.
- Mining rewards.
- Monetary supply.
- Consensus rules.

These rules must be enforced by the blockchain protocol and independently verified by nodes.

No application may bypass protocol validation.

END OF DOCUMENT